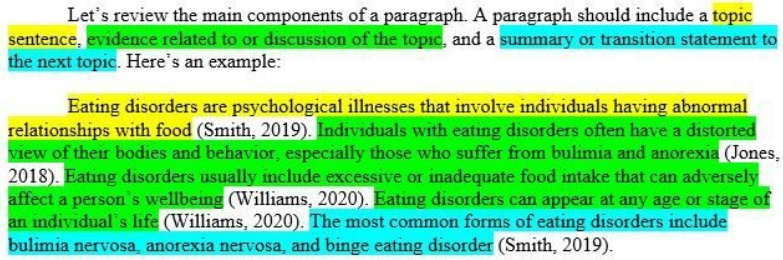
Dynamic collaborative filtering Thompson Sampling for cross-domain advertisements recommendation

Paper Review

Saturday, January 18, 2025



Let's review the main components of a paragraph. A paragraph should include a topic sentence, evidence related to or discussion of the topic, and a summary or transition statement to the next topic. Here's an example:

Eating disorders are psychological illnesses that involve individuals having abnormal relationships with food (Smith, 2019). Individuals with eating disorders often have a distorted view of their bodies and behavior, especially those who suffer from bulimia and anorexia (Jones, 2018). Eating disorders usually include excessive or inadequate food intake that can adversely affect a person's wellbeing (Williams, 2020). Eating disorders can appear at any age or stage of an individual's life (Williams, 2020). The most common forms of eating disorders include bulimia nervosa, anorexia nervosa, and binge eating disorder (Smith, 2019).

Figure 1: litrature review

https://youtu.be/DAL7cpE3K7E

So I don't have much time for this today so here is a quick note on: (Ishikawa et al. 2022)

TL;DR - Dynamic collaborative filtering via Thompson Sampling

Figure 2: DCTS in a nutshell

1. The authors are using **Thompson Sampling**. This is a Bayesian method in RL.
2. Thier problem is an advert recommendation system. So they are integrating Thompson sampling into making recommendations.
3. The talk mentions a dataset the authors used for doing this work. Is this dataset available? I would like to try this out

One line on Thompson sampling, one of the oldest technique in the RL playbook which uses the following rule: pick an action at random from the posterior distribution of the action values and then use the outcome to update the posterior distribution for the next step.

My ideas

- Find what data set was used.
  - Rakuten Ichiba data
  - Rakuten Travel dataset
- Is this dataset available?
- Can we make a minimal version to quickly test this kind of agent?

- Figure out a framework that extends tompson sampling to other RL problems.
  - need to add P(action|state) i.e. add conditioning of the bernulli on the state.
  - prehaps do simple counts of steps since starts or last reward.
  - prehaps using a succeror representation can help
- Marketing are the worst POMDPs. Testing real stuff is very hard so a good environment might help.
- I want to make an petting zoo env to support single & multiagent:
  1. auctions / non autions
  2. advertising (rec sys) with costs
  3. pricing with policies.
  - It should also allow incorperating real data from a dataset. Diretly or via sampling
  - It would be even neater to do this using a heirarchiacal model.
  - It would be even better if we can also incorportate the product, user hierecies.

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ABSTRACT

Recently online advertisers utilize Recommender systems (RSs) for display advertising to improve users' engagement. Contextual bandit model is a widely used RS to exploit and explore users' engagement and maximize the long-term rewards such as clicks or conversions. However, the current models aim to optimize a set of ads only in a specific domain and do not share information with other models in multiple domains. In this paper, we propose dynamic collaborative filtering Thompson Sampling (DCTS), the novel yet simple model to transfer knowledge among multiple bandit models. DCTS exploits similarities between users and between ads to estimate a prior distribution of Thompson sampling. Such similarities are obtained based on contextual features of users and ads. Similarities enable models in a domain that didn't have much data to converge more quickly by transferring knowledge. Moreover, DCTS incorporates temporal dynamics of users to track the user's recent change of preference. We first show transferring knowledge and incorporating temporal dynamics improve the performance of the baseline models on a synthetic dataset. Then we conduct an empirical analysis on a real-world dataset and the result showed that DCTS improves click-through rate by 9.7% than the state-of-the-art models. We also analyze hyper-parameters that adjust temporal dynamics and similarities and show the best parameter which maximizes CTR.

1 INTRODUCTION

Recently online advertisers utilize Recommender systems (RSs) for display advertising to improve users' engagement. Traditionally, collaborative filtering [1] is one of the popular recommender systems [2]. However, it suffers from the cold start problem [3]; Collaborative filtering cannot make a good recommendation for new items and new users without historical interactions. Content-based methods and hybrid models [4],[5] leverage contextual features (i.e., side information) of items and users to overcome the cold-start problem.

Meanwhile, reinforcement learning (RL) -based methods have shown its effectiveness on Recommender systems. By utilizing both exploration and exploitation, it can maximize the long-term profits for RS. It is suited for advertisement recommendation, where the

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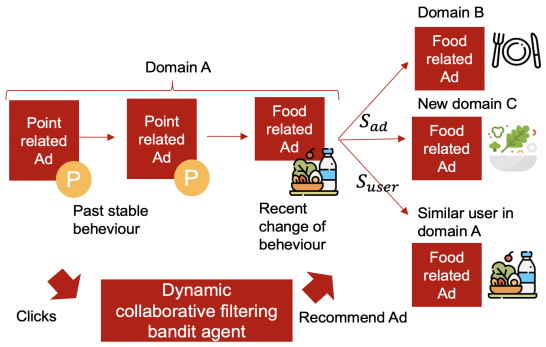

Figure 1: Use-case of the dynamic collaborative filtering ba dit.

ads are shown in the limited time period. Among reinforceme learning-based methods, bandit models were frequently applied that problem [6], which dynamically selects arms (correspondi to actions in RL) and maximizes total rewards. Contextual band that utilizes side-information of users and ads as contexts (e. user demography, creative characteristics) achieved great succe in various areas such as movie and book recommendation. (Li UCB [7] and LinTS [8]). Ads, unlike movies and products, are cr ated when new sales campaigns begin and are removed from d plays when campaigns are over. Therefore, ratios of newly creat ads are higher compared to new movies or products. Tradition methods such as collaborative filtering can't perform well for a recommendation because of the sparsity of training data and is not suited for long-term profit optimization. RL solved lon term profit problem and can deal with cold-start cases relative well, but it still suffers from this extreme cold-start conditions. this scenario, the agent tends to focus on exploration and sacrifi short-term reward for long-term reward. The ϵ-first policy [9], f instance, purely explores in cold-start conditions via random re ommendations, which goes overboard with the exploitation ar exploration balancing and results in even worse performance. He previous models were mainly created to optimize a specific widg on a web domain and each model was independent of each oth Considering that the number of active users in a domain (a busine unit) varies in large companies with multiple business units, it natural to transfer model's knowledge from popular domain unpopular/new domains to improve the recommendation perfo mance and convergence speed. Thus, in this paper we propose model to improve performance of existing RL-based recomme

Ishikawa, Shion, Young-joo Chung, and Yu Hirate. 2022. *Dynamic Collaborative Filtering Thompson Sampling for Cross-Domain Advertisements Recommendation*. https://arxiv.org/abs/2208.11926.